Rahul Rawat

MASTER THESIS

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PROFILE

Data Science and Analytics Master student at SRH Heidelberg with practical experience integrating external signals into production data pipelines and building leakage aware ML models. My cloud pipeline projects have covered live API ingestion, medallion architecture in BigQuery, and BigQuery ML classifiers with strict feature separation, all of which map directly onto the pricing prototype brief. I care about honest evaluation, backtesting, and documenting trade offs, which is what a Data and AI pricing team needs from a thesis student.

EXPERIENCE

eRay GmbH
Oct 2025 to Mar 2026

Data Scientist

- Built an end to end recursive time series pipeline forecasting chlorophyll a, turbidity, pH, and dissolved oxygen for a German lake, delivered as a six month collaboration with SRH University Heidelberg.
- Benchmarked six models head to head, Ridge, Gradient Boosting, LightGBM, XGBoost, CatBoost, and Prophet, and used CatBoost multi quantile regression to produce asymmetric 80 percent prediction intervals.
- Enforced strict anti leakage rules across the pipeline, surfacing the honest finding that pH and dissolved oxygen are physically predictable while chlorophyll a and turbidity are not without live optical sensors.
- Reconstructed missing winter readings with MICE imputation and engineered a synthetic winter decay forecast canvas so tree based models stopped flatlining during recursive prediction.
- Wrapped the whole pipeline in an orchestrator with gate checks and velocity and ecological bounds that halts on failed imputation rather than letting bad data cascade downstream.

EDUCATION

M.Sc. Data Science and Analytics
Apr 2025 to Present

SRH University of Applied Sciences Heidelberg

Bachelor of Technology in Computer Science
2019 to 2023

GL Bajaj Institute of Technology and Management

PERSONAL PROJECTS

Real Time Flight Tracking Data Pipeline

Python · PySpark · BigQuery · dbt · Apache Airflow · GCP Dataproc · GCE · GCS · Tableau · TabPy · OAuth2

- Set up real time data collection from the OpenSky Network API, polling live flight positions every 30 seconds and enriching each observation against airport, aircraft, and weather data from four external sources.
- Cleaned raw data with PySpark on Google Cloud and shaped it into analysis ready tables with dbt, computing each aircraft's nearest airport along the way and yielding more than 128 thousand joined records.
- Orchestrated the whole system with Apache Airflow so that the batch and real time layers refresh automatically every 15 minutes with no manual intervention.

- Surfaced findings in Tableau with Python statistics served through TabPy, showing that air traffic dropped by a factor of 4.4 in heavy rain and clustered heavily around hubs such as Frankfurt and Munich.

Movie Analytics and ML Pipeline, Cloud Native Data Platform

GCP BigQuery | Cloud Run | GCS | Cloud Scheduler | BigQuery ML | Python | SQL | Looker Studio

- Built an end to end batch pipeline pulling movie data from a public API into a GCS data lake, processed through a Bronze to Silver to Gold medallion architecture in BigQuery, and fully automated on a Cloud Scheduler trigger with no manual steps.
- Engineered the Silver layer for trustworthy analytics with schema enforcement, safe type casting, deduplication through window functions, and genre normalisation into a relational model.
- Trained a BigQuery ML classifier to predict whether a film will be a hit before release, deliberately splitting features into two tables to prevent data leakage so that only pre release signals feed the model.
- Built Gold layer aggregates and a five page Looker Studio dashboard answering concrete business questions on genre ROI, foreign language growth, and release season timing, plus an ML early warning view.
- Secured the system with a least privilege service account and Secret Manager and version controlled all code, SQL, and schemas in a reproducible GitHub repository.

CreditIQ, Fairness by Design Credit Scoring System

Python | scikit learn | AIF360 | SHAP | Streamlit

- Lifted the model's Disparate Impact ratio from a failing 0.79 to a compliant 0.88, clearing the EU AI Act and AGG 80 percent fairness threshold.
- Diagnosed a hidden intersectional bias where younger women were still being penalised even after single axis age bias was fixed, using SHAP, then designed a four way age by gender threshold matrix to correct it without over correcting into reverse discrimination.
- Cut the false negative rate from 44 percent to 16.7 percent while holding accuracy at 75 percent, documenting the fairness accuracy trade off as a deliberate and regulator defensible decision.
- Shipped a Streamlit decision support tool giving finance managers a recommendation plus a plain language LLM generated explanation, keeping a human in the loop per GDPR Article 22 and EU AI Act Article 14.
- Backed the pipeline with unit tests at 100 percent branch coverage and a full regulatory write up covering EU AI Act Annex III, GDPR, a model card, and attack vectors.

RESEARCH AND THESIS

Bachelor Thesis, Diabetes Prediction Using Machine Learning

GL Bajaj Institute of Technology and Management, IEEE style paper

- Built a full end to end machine learning pipeline comparing six classifiers on a clinical diabetes dataset of 768 patients, using 10 fold cross validation and per model confusion matrices.
- Caught biologically impossible zero values that the original authors had overlooked, then applied IQR based outlier removal and proper imputation to restore data integrity.
- Chose ROC AUC over accuracy as the headline metric for a 65 to 35 imbalanced dataset, avoiding a misleadingly rosy accuracy figure that would have hidden real error patterns.
- Wrote up findings as an IEEE style paper including an honest section on limitations and what to do differently in a follow up study.

LANGUAGES

English: fluent, professional working proficiency. **German:** B1, in progress.